

KPI forecasting and monitoring diagnostics

Forecast evaluation, model health and anomaly alerts

A monitoring workflow for insurance-style operational KPIs, combining time-series forecasts with driver analysis, residual checks and explicit alert rules.

Methods	R; ARIMA; regression; rolling forecast evaluation; anomaly monitoring
Data	Synthetic monthly insurance KPIs; January 2019–December 2025
Revision	21 September 2026

1 Executive summary

This project develops an end-to-end KPI forecasting and monitoring workflow in R. The dataset is synthetic, but it is designed to resemble a monthly insurance and operational performance setting. It includes claims count, earned premium, lapse rate, loss ratio and an operating margin proxy, together with supporting drivers such as policy exposure, inflation, seasonality and economic stress.

The modelling framework has two parts. First, ARIMA models are fitted to each KPI and used for twelve-month forecasts with prediction intervals. Second, AIC-selected regression models are used to link KPI movements to business drivers. The regression component is included not only for prediction, but also to show how operational and financial movements can be explained in business terms.

The archived outputs include 36 one-step forecast comparisons per KPI, with MAE, RMSE, MAPE, bias and 95% interval coverage. Reported coverage ranges from 94.4% to 97.2%. The supplied evaluation code reuses a model fitted on the full sample, so these figures are retrospective diagnostics and require a fresh, leakage-free evaluation before being described as out-of-sample performance.

The diagnostics section checks residual behaviour using summary statistics and Ljung-Box tests. The loss-ratio model is shown visually as a representative diagnostic example because loss ratio is a key insurance performance KPI. The same diagnostic process is applied across all forecasted KPIs.

The monitoring layer combines current KPI movements, lagged rolling thresholds and forecast-interval breaches into review triggers. Claims count and operating margin are flagged by residual autocorrelation checks, demonstrating why monitoring must cover model quality as well as the KPI values themselves.

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2 Data and scope

The project uses a synthetic monthly dataset from January 2019 to December 2025, giving 84 observations. The data are constructed to represent a realistic monitoring environment rather than a purely random sample. The series include trend, seasonality, exposure effects, economic stress effects and occasional shock behaviour.

The five main KPIs are:

- **Claims Count:** a volume measure representing the number of claims in each month.
- **Earned Premium / Revenue:** a financial volume measure used both as an insurance premium measure and a broader revenue proxy.
- **Lapse Rate:** a policyholder behaviour KPI showing the proportion of business lapsing in a period.
- **Loss Ratio:** a core insurance performance KPI measuring claims cost relative to earned premium.
- **Operating Margin Proxy:** a finance-oriented profitability measure that responds to claim and expense pressure.

The dataset also includes modelling drivers such as policy count, inflation index, economic stress, month-of-year effects and lagged KPI values. This allows the project to support both forecasting and explanatory analytics.

Table 1. Dataset summary.

Metric	Value
Data start	2019-01-01
Data end	2025-12-01
Monthly observations	84
Average policy count	12,581
Average earned premium / revenue	1,258,208
Average claims count	463
Average loss ratio	44.64%
Average lapse rate	4.12%
Average operating margin proxy	32.63%

The time series plot shows the intended structure of the synthetic data. Earned premium has a clear upward trend, while claims count, loss ratio and lapse rate show more short-term volatility. The shock around 2023 is visible in claims count, loss ratio and operating margin, creating a realistic stress period for the monitoring system to detect.

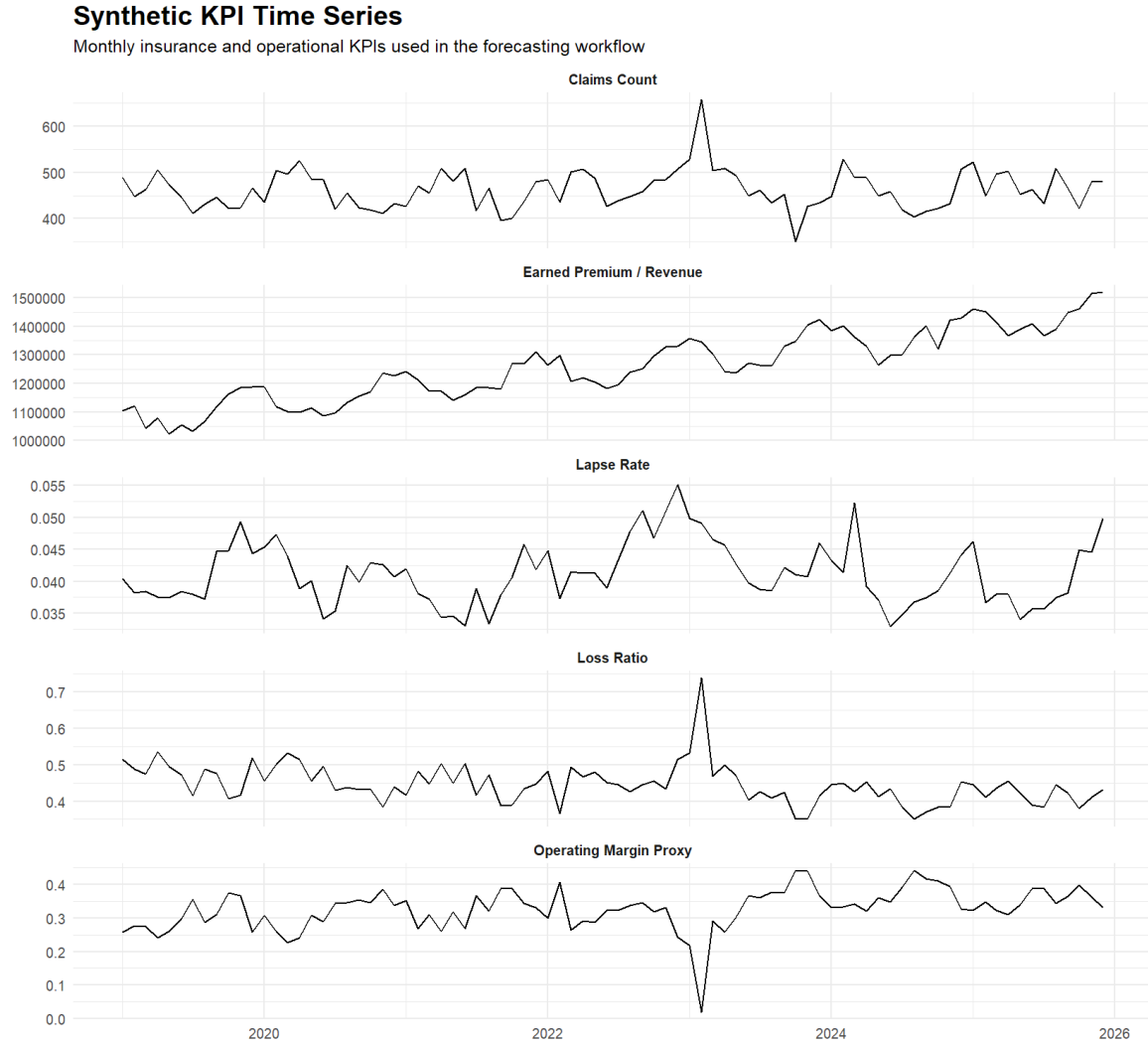


Figure 1. Synthetic monthly KPI series. The dataset combines insurance KPIs with broader operational and financial performance measures.

3 Methodology

3.1 ARIMA forecasting

Each KPI is modelled using an ARIMA specification. ARIMA models are appropriate as a baseline because they can capture autoregressive structure, differencing behaviour, seasonality and moving-average effects in a univariate time series. For a KPI series y_t , a simplified ARIMA structure can be written as:

$$\phi(B)(1 - B)^d y_t = c + \theta(B)\varepsilon_t, \quad (1)$$

where B is the backshift operator, d is the order of differencing, $\phi(B)$ captures autoregressive terms, $\theta(B)$ captures moving-average terms and ε_t is the residual innovation.

The automatic ARIMA search uses the small-sample corrected criterion AICc, the default in the supplied call to the R `forecast` package [6]. The model table reports AIC and BIC. The underlying AIC is:

$$\text{AIC} = 2k - 2\log(\hat{L}), \quad (2)$$

where k is the number of estimated parameters and $\hat{L}$ is the maximised likelihood. AIC balances goodness of fit against model complexity, so it is useful for selecting parsimonious time-series specifications [1].

Table 2. Selected ARIMA models.

KPI	ARIMA form	AIC	BIC	Innovation variance
Claims count	(1,0,0)	853.88	861.17	1,447
Earned premium / revenue	(0,0,0)(1,1,0)[12]	1670.75	1677.58	597,351,798
Lapse rate	(3,0,0)(0,1,2)[12]	-596.74	-583.08	0.00001
Loss ratio	(2,1,1)	-257.10	-247.42	0.00245
Operating margin proxy	(2,1,1)	-251.65	-241.97	0.00262

Forms are $(p, d, q)(P, D, Q)[12]$ where seasonal terms are present. Variances use the original KPI scale.

The selected ARIMA models vary across KPIs, which is expected. Earned premium requires seasonal differencing, while loss ratio and operating margin are better represented by differenced ARIMA specifications. Claims count is represented by a simpler autoregressive model.

3.2 Regression driver models

Regression models are included to connect KPI movements to interpretable business drivers. For a KPI y_t , the general regression form is:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_p x_{p,t} + u_t, \quad (3)$$

where the x variables represent drivers such as month-of-year effects, policy count, economic stress, inflation and lagged KPI values.

Stepwise AIC selection is used for the regression specifications. It balances fit against model size, but the selected relationships remain descriptive and should not be interpreted as causal effects.

The operating-margin model is handled carefully. Same-period loss ratio and same-period expense ratio are not used as inputs because those variables mechanically define operating margin. Including them would make the regression look artificially strong and would amount to reconstructing a formula rather than modelling a predictive relationship. Instead, the selected model uses lagged and external drivers.

The regression results are plausible for a synthetic business dataset. Earned premium is strongly explained by exposure and inflation drivers. Lapse rate is linked to economic stress and its own lag. Operating margin is explained with a moderate adjusted R^2 , which is more credible than a mechanically perfect fit.

Table 3. Selected regression models.

KPI	Selected drivers	AIC	Adjusted R^2	Residual SE
Claims count	Month effects, policy count and economic stress	818.68	0.476	30.70724
Earned premium / revenue	Month effects, policy count, economic stress and inflation index	1898.28	0.972	20,402
Lapse rate	Month effects, economic stress and lagged lapse rate	-734.01	0.704	0.00266
Loss ratio	Month effects, economic stress and inflation index	-284.93	0.474	0.03981
Operating margin proxy	Month effects, economic stress, inflation index, lagged loss ratio and lagged operating margin	-287.59	0.594	0.03881

3.3 Rolling forecast evaluation

The evaluation output contains 36 one-step comparisons, beginning after an initial 48 observations. The supplied code applies `Arima(train_ts, model = fit)`, where `fit` was estimated on the full series. This call reuses fitted parameters rather than estimating them afresh [7]. Although each forecast receives an expanding historical series, full-sample parameter information enters the calculation. The reported metrics are therefore retained as retrospective diagnostics; a strict rolling evaluation must select and fit each model using only observations available at that forecast origin.

The main error metrics are:

$$\text{MAE} = \frac{1}{n} \sum_{t=1}^n |y_t - \hat{y}_t|, \quad (4)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2}, \quad (5)$$

$$\text{MAPE} = \frac{100}{n} \sum_{t=1}^n \left| \frac{y_t - \hat{y}_t}{y_t} \right|. \quad (6)$$

Bias is calculated as the average forecast error. Prediction interval coverage is calculated as the proportion of actual observations falling inside the model's 95% prediction interval.

4 Results and discussion

4.1 Forecast outputs and evaluation

The forecast output is summarised using a compact chart showing historical values, twelve-month forecasts and 95% prediction intervals for all five KPIs.

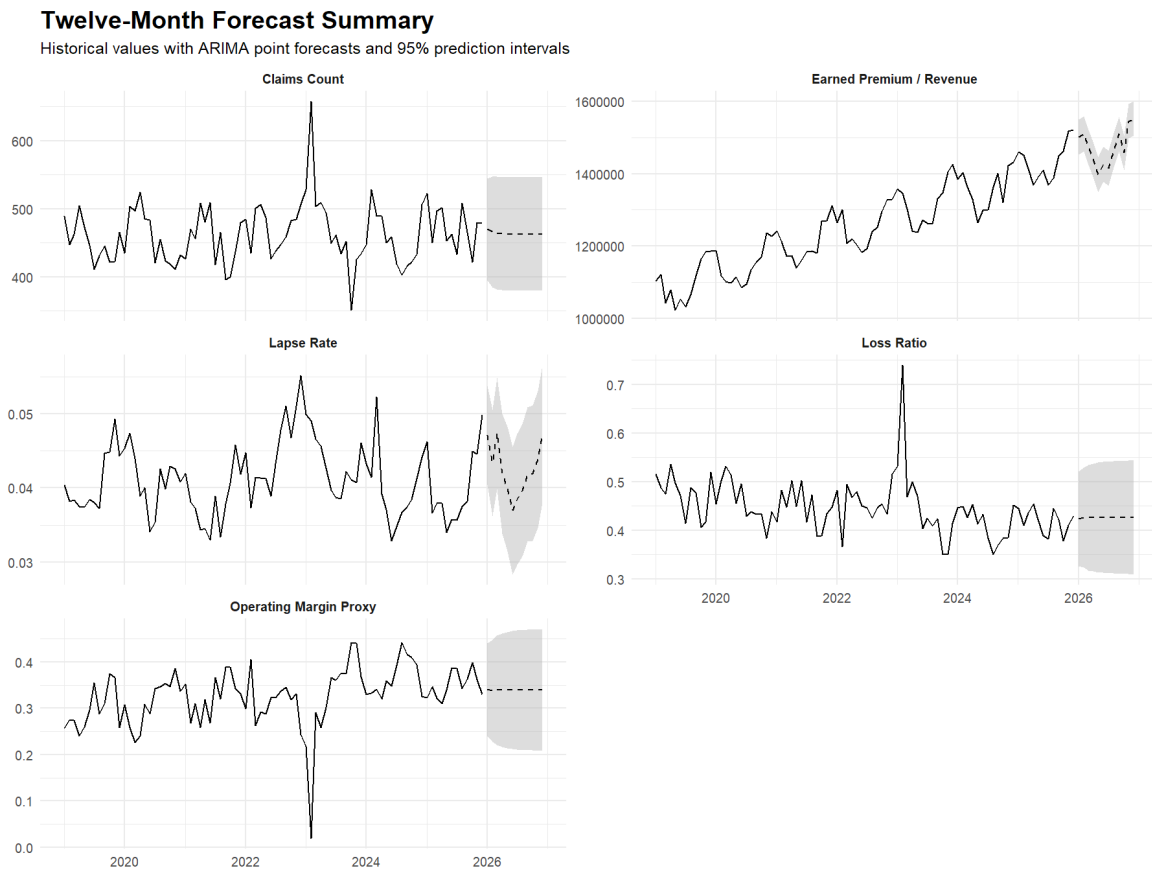


Figure 2. Twelve-month ARIMA forecast summary. Historical values are shown with point forecasts and 95% prediction intervals for each KPI.

The forecast profiles differ across the five KPIs. Earned premium continues its upward trend, while claims count remains more variable. Rate and margin forecasts need plausibility checks because an unconstrained ARIMA model does not itself enforce economically meaningful bounds.

Table 4. Reported one-step forecast diagnostics.

KPI	<i>n</i>	MAE	RMSE	MAPE	Bias	95% coverage
Claims count	36	32.4	45.2	6.8%	2.3	94.4%
Earned premium / revenue	36	19,718	26,326	1.5%	1,773	94.4%
Lapse rate	36	0.26	0.33	6.3%	-0.04	97.2%
Loss ratio	36	3.89	5.76	8.8%	-0.93	94.4%
Operating margin proxy	36	4.12	5.89	42.8%	0.99	94.4%

MAE, RMSE and bias use natural units for counts/revenue and percentage points for rates/margins. These archived diagnostics reuse full-sample fitted parameters and are not strict out-of-sample estimates.

For claims count and earned premium, MAE, RMSE and bias are reported in the natural units of the KPI. For lapse rate, loss ratio and operating margin, MAE, RMSE and bias are reported in percentage-point terms. MAPE remains a relative percentage error for all KPIs.

Observed interval coverage is close to 95%, but the full-sample fit reuse and the short evaluation period prevent a claim of independent calibration. Operating margin has high MAPE because

percentage errors are sensitive to small or unstable denominators. Its percentage-point MAE and RMSE are more directly interpretable.

4.2 Forecast error monitoring

Rolling forecast errors are monitored using a six-month rolling RMSE. This provides a practical way to identify periods where forecast quality deteriorates. A model can have acceptable average performance but still show periods of instability, so rolling diagnostics are useful for monitoring model risk over time.

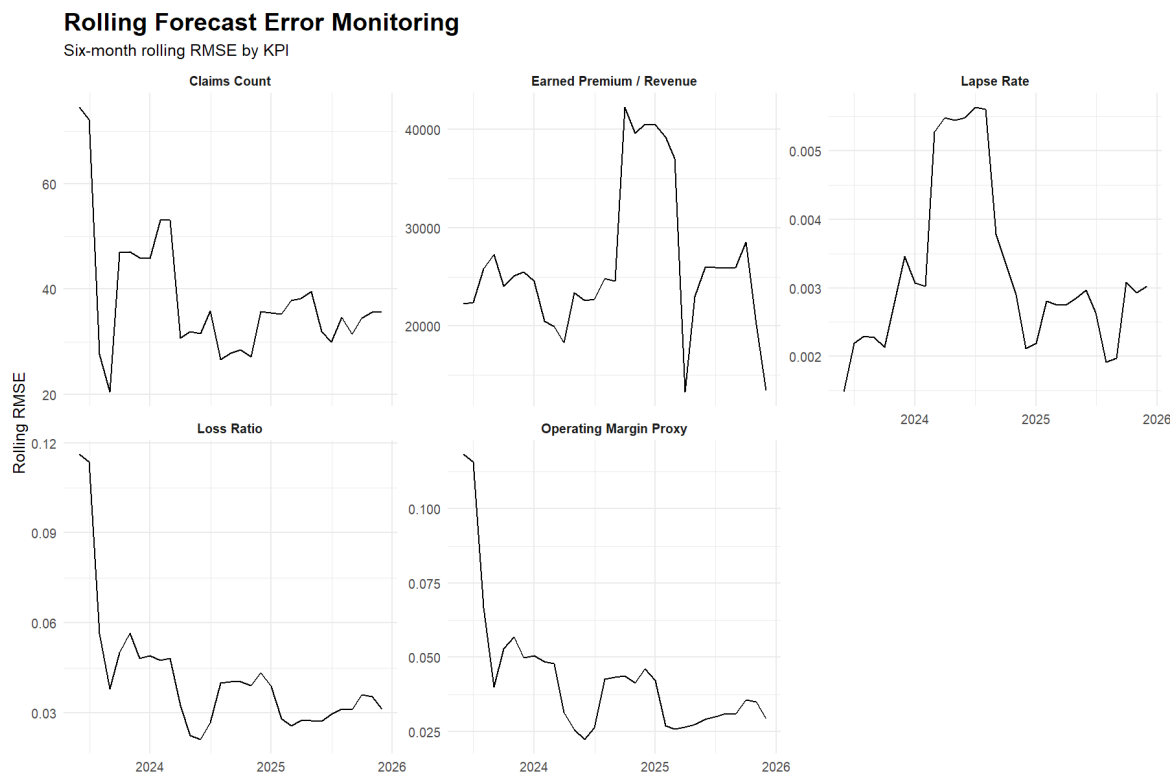


Figure 3. Six-month rolling RMSE by KPI. Rolling forecast errors help identify periods where model performance deteriorates or improves.

The rolling RMSE chart shows that forecast errors vary over time. Earned premium has a visible error spike around periods where the level of the series changes. Claims count and lapse rate also show periods of higher forecast error. This is useful from a monitoring perspective because forecast error instability can be treated as a review trigger rather than only as a model evaluation statistic.

4.3 Model health diagnostics

Residual diagnostics were used to check whether the fitted ARIMA models left behind material autocorrelation. The Ljung-Box test is used as a portmanteau test for residual autocorrelation [5]. A low p-value indicates that the residuals may still contain time-series structure not captured by the model.

The Ljung-Box results flag claims count and operating margin for review. Earned premium, lapse rate and loss ratio do not reject the no-autocorrelation null at the selected test settings. Non-

Table 5. Model health summary.

KPI	Ljung–Box p-value	Residual mean	Residual SD	Test outcome
Claims count	0.0014	-0.2	37.8	Review
Earned premium / revenue	0.4692	-381	22,442	Non-rejection
Lapse rate	0.3678	0.01	0.30	Non-rejection
Loss ratio	0.2253	-0.76	4.80	Non-rejection
Operating margin proxy	0.0063	0.73	4.97	Review

ARIMA forms are given in the selected-model table. Residual mean and SD use natural units for counts/revenue and percentage points for rates/margins. Non-rejection refers to the 5% test level, not proof of independent residuals.

rejection is not proof that the residuals are independent; the diagnostics identify where additional structure is most evident.

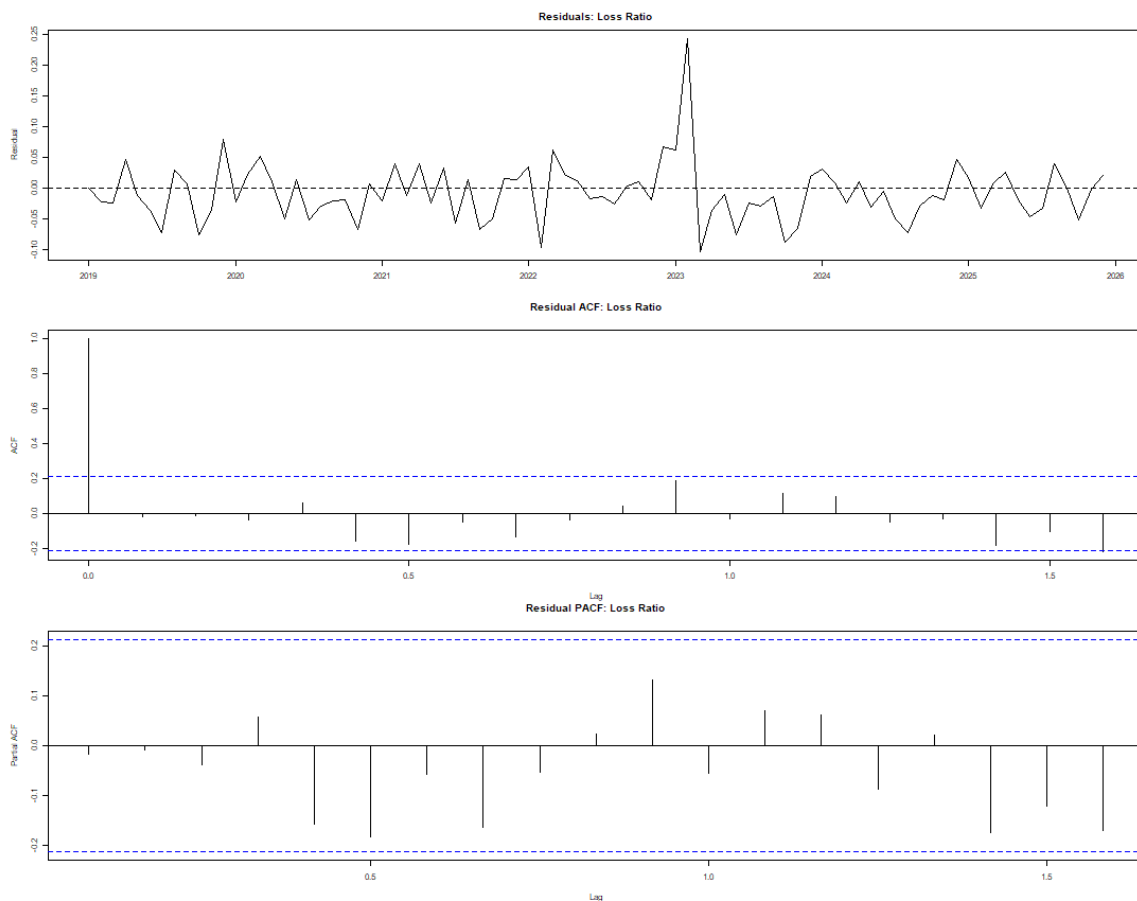


Figure 4. Representative loss-ratio residual diagnostics. The same residual summary process is applied across all five forecasted KPIs, while loss ratio is shown visually because it is a key insurance performance measure.

The loss-ratio plots illustrate the residual, ACF and PACF checks, while the table reports the corresponding model-health results for all five KPIs.

4.4 KPI movements and anomaly alerts

The monitoring layer converts the modelling results into a dashboard-style process. It uses two main alert types:

- **Rolling threshold breach:** the current KPI value is compared with a prior 12-month rolling mean and a plus/minus two-standard-deviation band.
- **Forecast interval breach:** the actual value is compared with the one-step-ahead 95% prediction interval.

The rolling threshold uses prior observations only. This avoids look-ahead bias and makes the alert rule closer to a process that could have been applied at the time.

4.4.1 Current KPI snapshot

The current KPI snapshot reports the final observation date and recent movement across the five KPIs. Volume and revenue measures are reported using relative percentage change. Rate and margin KPIs are reported using percentage-point change because relative percentage changes can be misleading for bounded ratio series.

Table 6. KPI snapshot at 1 December 2025.

KPI	Current value	MoM change	YoY change	Change basis
Claims count	479	-0.2%	-5.5%	Relative %
Earned premium / revenue	1,520,433	+0.3%	+6.3%	Relative %
Lapse rate	4.98%	+0.52	+0.57	Percentage points
Loss ratio	43.11%	+2.12	-2.13	Percentage points
Operating margin proxy	33.00%	-3.17	+0.38	Percentage points

MoM: month-on-month. YoY: year-on-year. The observation date is 2025-12-01 for every row.

The snapshot shows why a single movement definition is not appropriate for all KPI types. A 2 percentage-point change in loss ratio or operating margin can be commercially meaningful even if the relative percentage movement is less intuitive.

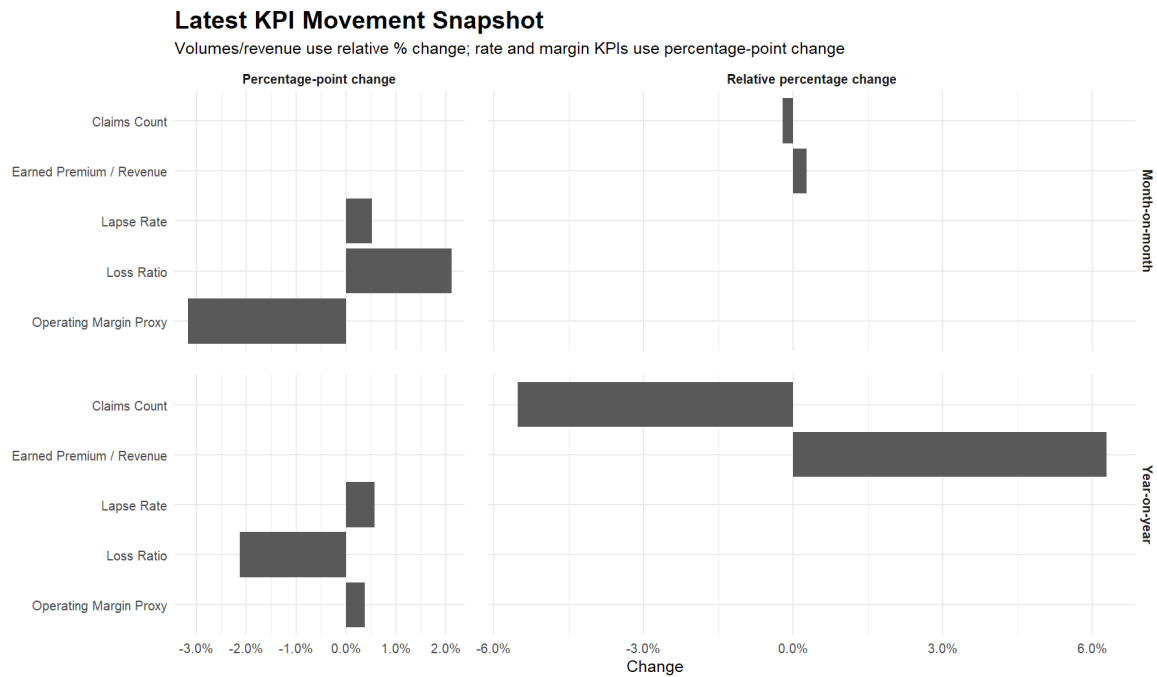


Figure 5. Latest KPI movement snapshot. Volumes and revenue use relative percentage change; rates and margins use percentage-point change.

4.4.2 Alert summary

The alert summary counts rolling threshold breaches and forecast interval breaches by KPI. This makes the monitoring output easier to read than a long raw alert table.

Table 7. Alert count summary by KPI and alert type.

KPI	Alert type	Alert count
Claims Count	Forecast interval breach	2
Claims Count	Rolling threshold breach	4
Earned Premium / Revenue	Forecast interval breach	2
Earned Premium / Revenue	Rolling threshold breach	8
Lapse Rate	Forecast interval breach	1
Lapse Rate	Rolling threshold breach	9
Loss Ratio	Forecast interval breach	2
Loss Ratio	Rolling threshold breach	4
Operating Margin Proxy	Forecast interval breach	2
Operating Margin Proxy	Rolling threshold breach	3

Earned premium and lapse rate have the highest number of rolling threshold alerts. This does not automatically mean the models are poor. It indicates that these KPIs experienced movements outside their recent rolling behaviour and would merit review under the monitoring rules.

The tables show the ten most recent alerts. The accompanying CSV retains the full alert history for review and follow-up.

Table 8. Recent earned premium / revenue alerts.

Date	Rule	Actual	Reference	Lower	Upper
2025-12-01	R, above	1,520,433	1,425,755	1,337,454	1,514,057
2025-11-01	R, above	1,516,358	1,417,931	1,350,485	1,485,376
2025-05-01	F, outside	1,390,540	1,339,196	1,291,293	1,387,099
2024-10-01	F, outside	1,319,876	1,404,779	1,356,876	1,452,682
2023-12-01	R, above	1,425,169	1,307,865	1,202,650	1,413,081
2023-11-01	R, above	1,404,575	1,301,517	1,214,071	1,388,962

R: prior 12-month rolling threshold breach. F: one-step forecast interval breach. Reference is the rolling mean for R and the point forecast for F. Lower and upper are the corresponding bounds.

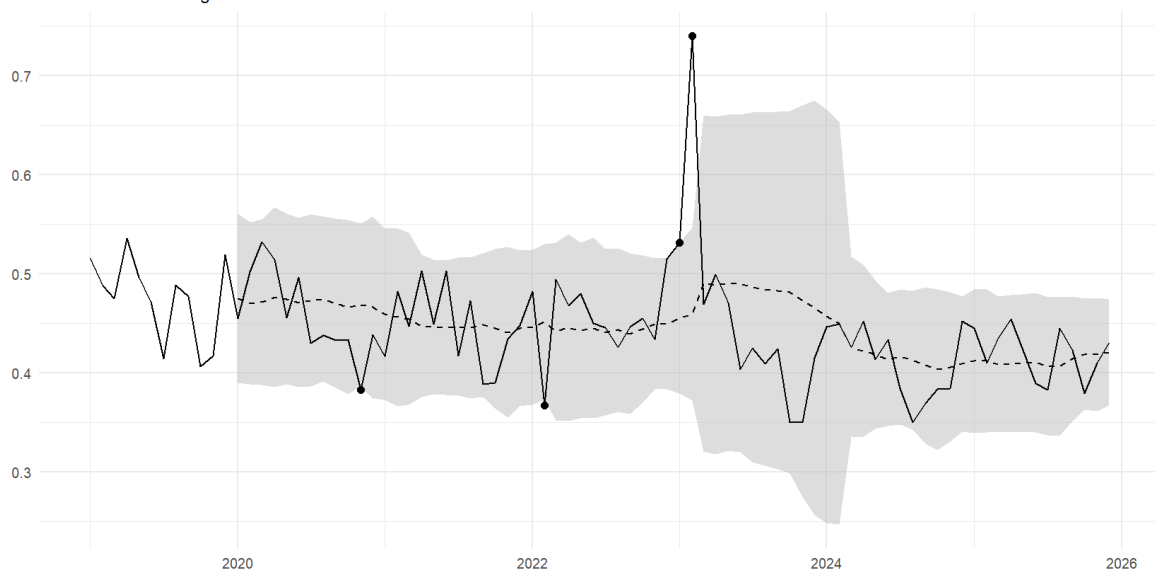
Table 9. Recent lapse rate alerts.

Date	Rule	Actual	Reference	Lower	Upper
2025-12-01	R, above	4.98%	3.94%	3.09%	4.80%
2024-06-01	R, below	3.29%	4.17%	3.35%	4.99%
2024-03-01	F, outside	5.23%	4.17%	3.50%	4.83%
2024-03-01	R, above	5.23%	4.22%	3.67%	4.77%

R and F follow the definitions in the preceding table. Together, the two tables contain the ten most recent alerts, grouped by KPI for readability.

Loss Ratio Monitoring Thresholds

Prior 12-month rolling mean with +/- 2 standard deviation bands

**Figure 6.** Loss-ratio monitoring against prior 12-month rolling threshold bands. The shaded band represents the rolling mean plus or minus two standard deviations.

The loss-ratio monitoring chart shows how the threshold rule behaves during stable and stressed periods. The band widens after the 2023 shock, which is expected because the prior 12-month standard deviation increases. This is useful from a monitoring perspective because it avoids treating every high value as equally unusual regardless of the recent volatility regime.

4.5 Operational interpretation

The workflow separates three questions: how the KPI is moving, how reliable the model is, and whether an observation merits investigation. ARIMA forecasts provide a baseline trajectory, regressions describe associations with business drivers, and threshold rules create consistent review triggers.

An alert is not a diagnosis of business deterioration. For example, higher earned premium can breach an upper threshold for favourable reasons, while a loss-ratio increase may require a different response. Operational follow-up should therefore consider the KPI's direction, materiality and business context alongside model health.

5 Limitations and further work

The project has several limitations.

- **Evaluation leakage:** the supplied rolling routine reuses a full-sample fitted model. Re-estimating parameters and selecting specifications within each training window is the first priority before making out-of-sample accuracy or calibration claims.
- **Synthetic data:** the dataset is designed to be realistic, but it is not a live insurance portfolio. The results should therefore be interpreted as a modelling demonstration rather than a business conclusion.
- **Baseline forecasting models:** ARIMA provides a useful benchmark, but a production system could also consider ARIMAX, dynamic regression, state-space models or hierarchical forecasting.
- **Residual autocorrelation:** claims count and operating margin are flagged for residual autocorrelation. In a production setting, these models would need further refinement.
- **MAPE limitations:** MAPE can be difficult to interpret for rate and margin series. Percentage-point MAE and RMSE are often more useful for these KPIs.
- **Threshold sensitivity:** the alert system uses a prior 12-month mean and plus/minus two standard deviation bands. Other businesses may prefer percentile-based, risk-based or governance-driven thresholds.

Further work could include adding exogenous ARIMAX forecasts, automated recalibration rules, model version tracking, severity-based alert ranking and a more formal model governance section. For insurance use cases, the project could also be extended to include claims severity, combined ratio, reserve development or pricing adequacy indicators.

The reported rolling metrics concern the ARIMA component. Regression adjusted R^2 describes fit to the modelling sample and is not a separate forecast-validation result. With only 36 forecast comparisons per KPI, coverage estimates also have substantial sampling uncertainty.

6 Conclusion

The project brings forecasts, driver analysis, residual diagnostics and anomaly rules into one reporting workflow. It makes both KPI movements and model-health issues visible, with claims count and operating margin identified for further model review.

The current outputs demonstrate the monitoring design. The next technical priority is to rerun the rolling evaluation with model selection and estimation confined to each training window. That would provide a sound basis for assessing forecast accuracy, interval coverage and alert performance.

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